

Group8_assignment_2

2026-07-08

(a)

```
# Group 8 assignment
# Calling my packages
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.1      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2     4.0.3      v tibble    3.3.1
## v lubridate  1.9.5      v tidyr     1.3.2
## v purrr      1.2.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

library(ggplot2)

# Working with the mtcars data set
## Loading the data set
Mtcars <- mtcars

## Displaying the first six observations
head(Mtcars)
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
## Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
## Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
## Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
## Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
## Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2
## Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0	3	1

- A brief description of the structure of the mtcars data set:
 - It has the name of the car model in the first column, followed by the mpg, cyl, disp, hp, drat, wt, qsec, vs, am, gear, and carb. They are all of the dbl data type.

(b)

(i)

```
# Performing descriptive analytics on the variable mpg  
mean(Mtcars$mpg) # mean
```

```
## [1] 20.09062
```

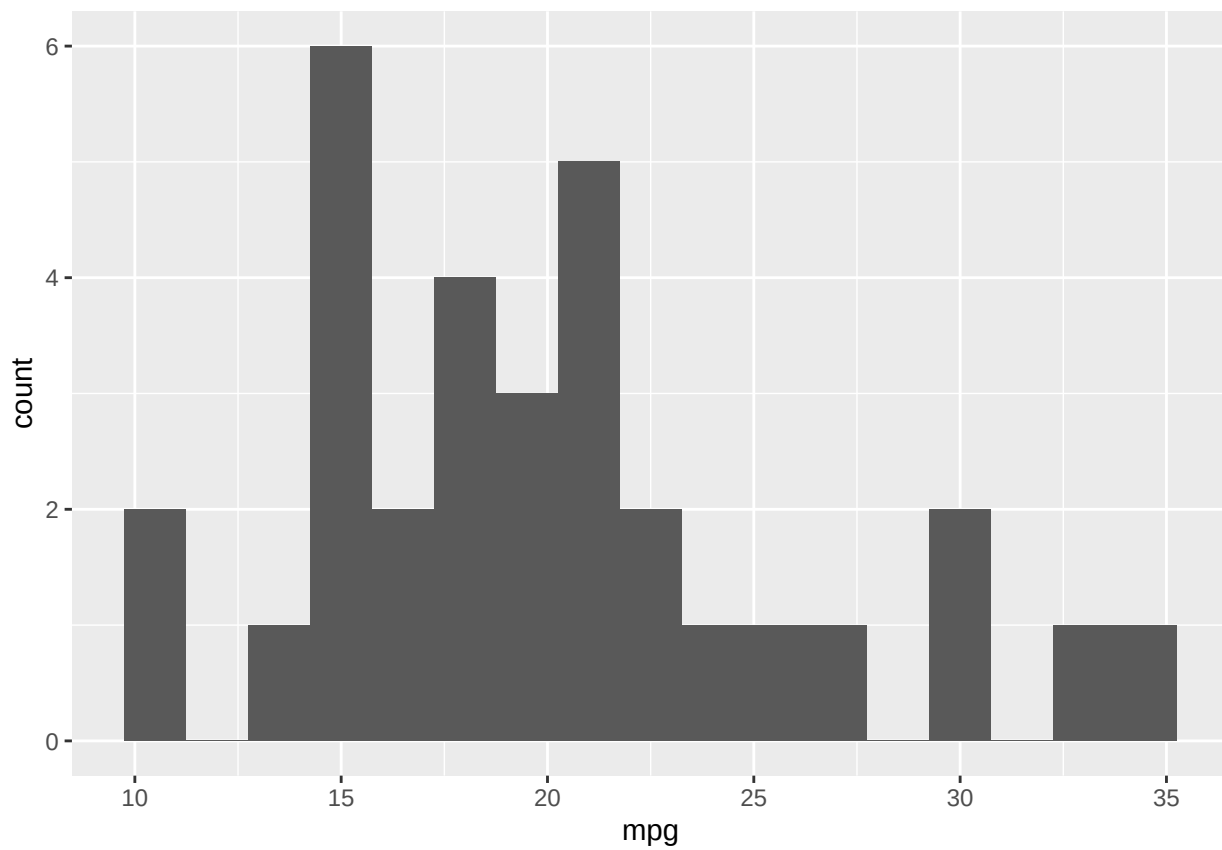
```
sd(Mtcars$mpg) # standard deviation
```

```
## [1] 6.026948
```

- Mean = 20.09062
- Standard deviation = 6.026948

(ii)

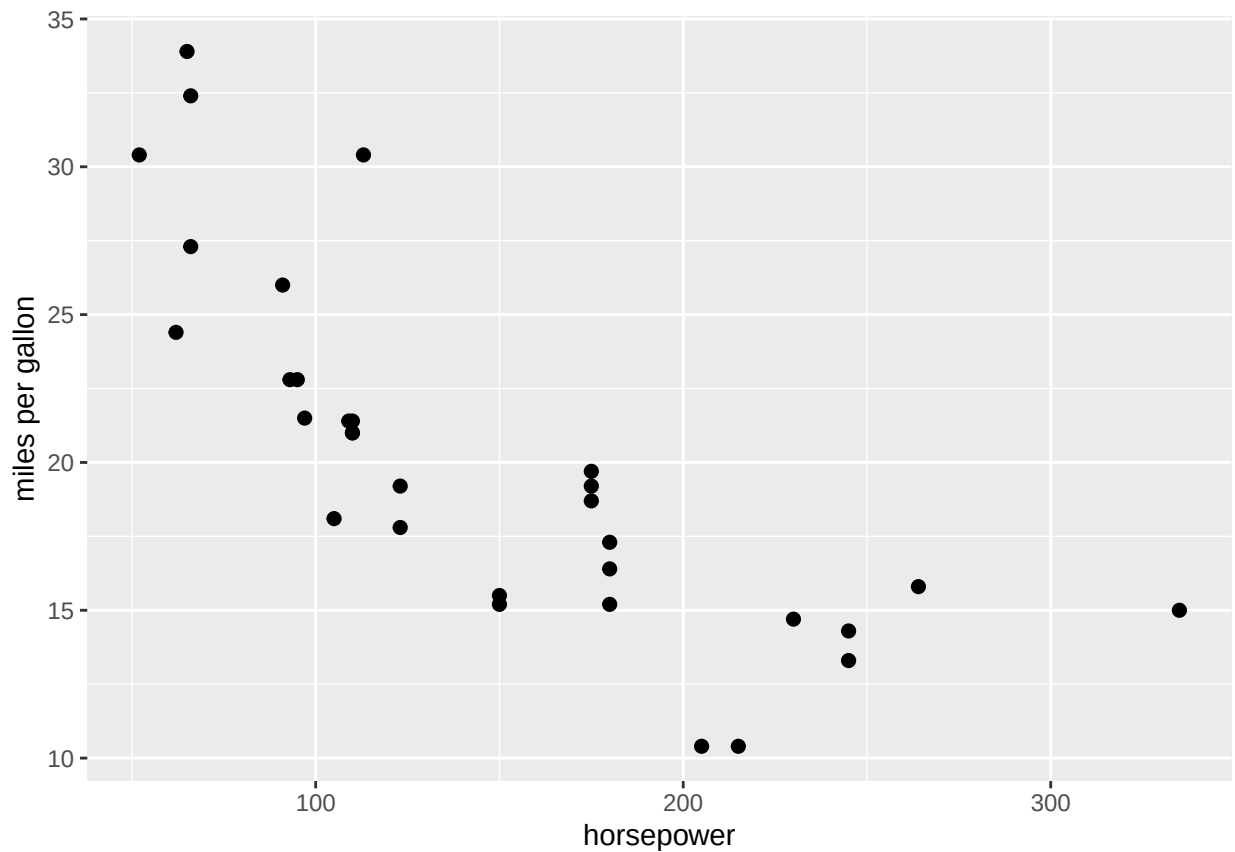
```
## Histogram for mpg  
ggplot2::ggplot(data = Mtcars, mapping = aes(x = mpg)) +  
  geom_histogram(binwidth = 1.5)
```



- Most of the data is clustered between approximately 12.5 and 27.5 miles per gallon

(c)

```
# Creating a scatter plot of mpg vs hp
ggplot2::ggplot(data = Mtcars, mapping = aes(x = hp, y = mpg)) +
  geom_point(size = 2) +
  labs (
    x = "horsepower",
    y = "miles per gallon"
  )
```



- There is a linear relationship

(d)

```
# Creating a simple linear regression model with mpg as the response variable and hp as the predictor variable
linear_model <- lm(mpg ~ hp, data = Mtcars)
summary(linear_model) # viewing the results of the linear model

##
## Call:
## lm(formula = mpg ~ hp, data = Mtcars)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.7121 -2.1122 -0.8854  1.5819  8.2360
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 30.09886    1.63392  18.421  < 2e-16 ***
## hp          -0.06823    0.01012  -6.742 1.79e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.863 on 30 degrees of freedom
## Multiple R-squared:  0.6024, Adjusted R-squared:  0.5892
## F-statistic: 45.46 on 1 and 30 DF,  p-value: 1.788e-07
```

(i)

- $y = 30.09886 - 0.06823x$

(ii)

- The coefficient implies that for every addition unit of horsepower, the miles per gallon reduce by 0.06823

(iii)

- The coefficient of determination = 0.6024
– or (60.24%)

(e)

```
# Predicting the fuel efficiency for a car with hp = 150
predict(linear_model, data.frame(hp = c(150)))
```

```
##      1
## 19.86462
```

- For a car with hp = 150 the linear model predicts an mpg of 19.86462